

Safety data sheet

Page: 1/14

BASF Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)

Date / Revised: 06.10.2023

Version: 1.0

Product: **Plantatex® LLE Plus**

(ID no. 30843601/SDS_GEN_00/EN)

Date of print 08.02.2024

1. Identification

Product identifier

Plantatex® LLE Plus

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Raw material for the chemical-technical industry

Details of the supplier of the safety data sheet

Company:

BASF SE

67056 Ludwigshafen

GERMANY

Operating Division Care Chemicals

Telephone: +49 621 60-57579

E-mail address: emd-ems-ehs-masterdata@basf.com

Emergency telephone number

International emergency number:

Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Eye Dam./Irrit. 1

Aquatic Acute 3

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:



Signal Word:

Danger

Hazard Statement:

H318 Causes serious eye damage.

H402 Harmful to aquatic life.

Precautionary Statements (Prevention):

P280 Wear eye and face protection.

P273 Avoid release to the environment.

Precautionary Statements (Response):

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P310 Immediately call a POISON CENTER or physician.

Precautionary Statements (Disposal):

P501 Dispose of contents and container to hazardous or special waste collection point.

Labeling of special preparations (GHS):

This surfactant complies with the biodegradability criteria as laid down in Regulation (EC) No.648/2004 on detergents. Data to support this assertion are held at the disposal of the competent authorities of the Member States and will be made available to them at their direct request or at the request of a detergent manufacturer.

Other hazards

According to UN GHS criteria

No specific dangers known, if the regulations/notes for storage and handling are considered.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

Aqueous solution based on: Alkylpolyglucoside, Glycerides, C16-18 and C18-unsatd. mono- and di-, Citric acid

Hazardous ingredients (GHS)

According to UN GHS criteria

D-Glucopyranose, oligomeric, C10-16-alkyl glycosides

Content (W/W): $\geq 15\%$ - $< 25\%$	Skin Corr./Irrit. 2
CAS Number: 110615-47-9	Eye Dam./Irrit. 1
	Aquatic Acute 2
	H318, H315, H401

Specific concentration limit:

Eye Dam./Irrit. 1: $> 12\%$
Skin Corr./Irrit. 2: $> 30\%$

D-Glucopyranose, oligomers, decyl octyl glycosides

Content (W/W): $\geq 10\%$ - $< 15\%$	Eye Dam./Irrit. 1
CAS Number: 68515-73-1	Aquatic Acute 3
EC-Number: 500-220-1	H318, H402

Citric acid

Content (W/W): $\geq 3\%$ - $< 7\%$	Acute Tox. 5 (oral)
CAS Number: 77-92-9	Eye Dam./Irrit. 2A
EC-Number: 201-069-1	STOT SE 3 (irr. to respiratory syst.)
	H319, H303, H335

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:

Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:

Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)

Date / Revised: 06.10.2023

Version: 1.0

Product: **Plantatex® LLE Plus**

(ID no. 30843601/SDS_GEN_00/EN)

Date of print 08.02.2024

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Hazards: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures**Extinguishing media**

Suitable extinguishing media:
water spray, dry powder, foam

Special hazards arising from the substance or mixture

harmful vapours, carbon oxides

Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:

Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures

High risk of slipping due to leakage/spillage of product.

Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Information regarding personal protective measures, see section 8.
Avoid contact with eyes.

Environmental precautions

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

For large amounts: Dike spillage. Pump off product.

For residues: Pick up with suitable absorbent material.

Dispose of absorbed material in accordance with regulations.

7. Handling and Storage**Precautions for safe handling**

No eating, drinking, smoking or tobacco use at the place of work. Wash hands before breaks and at end of work. Remove contaminated clothing and protective equipment before entering eating areas. Avoid contact with the skin, eyes and clothing. Use suitable eye protection.

Protection against fire and explosion:

No special precautions necessary.

Conditions for safe storage, including any incompatibilities

Suitable materials for containers: High density polyethylene (HDPE), Stainless steel 1.4306 (V2A), Stainless steel 1.4541, Stainless steel 1.4571

Further information on storage conditions: Keep container tightly closed and in a cool place.

Storage stability:

Storage temperature: 10 - 40 °C

Avoid freezing.

Protect from temperatures below: 10 °C

Below the temperature limit, the product is no longer pumpable.

Protect from temperatures above: 40 °C

Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

57-55-6: Propane-1,2-diol

77-92-9: Citric acid

Exposure controls

Personal protective equipment

Respiratory protection:

Respiratory protection in case of vapour/aerosol release. Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:

Chemical resistant protective gloves (EN ISO 374-1)

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):

nitrile rubber (NBR) - 0.4 mm coating thickness

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Tightly fitting safety goggles (cage goggles) (e.g. EN 166) and face shield.

Body protection:

Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)

Date / Revised: 06.10.2023

Version: 1.0

Product: **Plantatex® LLE Plus**

(ID no. 30843601/SDS_GEN_00/EN)

Date of print 08.02.2024

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Wearing of closed work clothing is recommended. No eating, drinking, smoking or tobacco use at the place of work. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	liquid	
Colour:	yellow	
Odour:	slight odour	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
pH value:	approx. 3,9 (water, 5 %(m)) (as an emulsion)	(internal method)
Freezing point:	not determined	
boiling temperature:	> 100 °C	(estimated)
Flash point:	> 101 °C	(DIN 51755)
Evaporation rate:	not determined	
Flammability:	hardly combustible	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Ignition temperature:	not determined	
Vapour pressure:	approx. 23 hPa (20 °C)	(estimated)
Density:	contains water approx. 1,091 g/cm3 (20 °C)	(DIN 51757)
	approx. 1,077 g/cm3 (55 °C)	(DIN 51757)
Relative density:	No data available.	
Relative vapour density (air):	not determined	
Solubility in water:	soluble 20 g/l (25 °C)	

Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)

Date / Revised: 06.10.2023

Version: 1.0

Product: **Plantatex® LLE Plus**

(ID no. 30843601/SDS_GEN_00/EN)

Date of print 08.02.2024

Partitioning coefficient n-octanol/water (log Kow):
Study does not need to be conducted.

Self ignition: not self-igniting

Thermal decomposition: No decomposition if stored and handled as prescribed/indicated.

Viscosity, dynamic: approx. 2.700 mPa.s (DIN EN ISO 2555 (LV))
(23 °C)

Viscosity, kinematic: not determined

Explosion hazard: not explosive

Fire promoting properties: not fire-propagating

Other information

Self heating ability: It is not a substance capable of spontaneous heating.

Hygroscopy: Non-hygroscopic

Surface tension: No data available.

Grain size distribution: The substance / product is marketed or used in a non solid or granular form.

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity**Reactivity**

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: No corrosive effect on metal.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

No hazardous reactions when stored and handled according to instructions.

Conditions to avoid

See SDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
acids, Alkalines, caustics, halogens, reactive chemicals

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion.

Experimental/calculated data:
ATE rat (oral): > 2.000 mg/kg (calculated)

LC50 rat (by inhalation):
No data available.

LD50 rat (dermal):
No data available.

Information on: D-Glucopyranose, oligomeric, C10-16-alkyl glycosides

Experimental/calculated data:
LD50 rat (oral): > 5.000 mg/kg (similar to OECD guideline 401)

Information on: D-Glucopyranose, oligomers, decyl octyl glycosides

Experimental/calculated data:
LD50 rat (oral): > 2.000 mg/kg (similar to OECD guideline 401)

Information on: Citric acid

Experimental/calculated data:
LD50 rat (oral): 3.000 mg/kg (OECD Guideline 401)

LD50 mouse (oral): 5.400 mg/kg (similar to OECD guideline 401)

Irritation

Assessment of irritating effects:
May cause severe damage to the eyes. Not irritating to the skin.

Information on: D-Glucopyranose, oligomeric, C10-16-alkyl glycosides
Experimental/calculated data:
Serious eye damage/irritation rabbit: irreversible damage (OECD Guideline 405)

Information on: D-Glucopyranose, oligomers, decyl octyl glycosides
Experimental/calculated data:
Serious eye damage/irritation rabbit: irreversible damage (OECD Guideline 405)
Data of a comparable product

Respiratory/Skin sensitization

Assessment of sensitization:

Based on the ingredients, there is no suspicion of a skin-sensitizing potential.

Germ cell mutagenicity**Assessment of mutagenicity:**

Based on the ingredients, there is no suspicion of a mutagenic effect.

Carcinogenicity**Assessment of carcinogenicity:**

Based on the ingredients there is no suspicion of a carcinogenic effect in humans.

Reproductive toxicity**Assessment of reproduction toxicity:**

Based on the ingredients, there is no suspicion of a toxic effect on reproduction.

Developmental toxicity**Assessment of teratogenicity:**

Based on the ingredients, there is no suspicion of a teratogenic effect.

Specific target organ toxicity (single exposure)**Assessment of STOT single:**

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)**Assessment of repeated dose toxicity:**

Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:

Acutely harmful for aquatic organisms.

Information on: D-Glucopyranose, oligomeric, C10-16-alkyl glycosides

Toxicity to fish:

Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)

Date / Revised: 06.10.2023

Version: 1.0

Product: **Plantatex® LLE Plus**

(ID no. 30843601/SDS_GEN_00/EN)

Date of print 08.02.2024

LC50 (96 h) 2,95 mg/L a.i., *Brachydanio rerio* (OECD 203; ISO 7346; 84/449/EWG, C.1, semistatic)
Nominal concentration.

Information on: D-Glucopyranose, oligomers, decyl octyl glycosides

Toxicity to fish:

LC50 (96 h) 126 mg/L a.i., *Brachydanio rerio* (OECD 203; ISO 7346; 84/449/EWG, C.1, semistatic)
Nominal concentration.

Information on: D-Glucopyranose, oligomeric, C10-16-alkyl glycosides

Aquatic invertebrates:

EC50 (48 h) 7 mg/L a.i., *Daphnia magna* (Directive 92/69/EEC, C.2, static)

The statement of the toxic effect relates to the analytically determined concentration.

LC50 (10 d) 650 mg a.i./kg dw, *Corophium volutator* (, static)

Nominal concentration.

Information on: D-Glucopyranose, oligomers, decyl octyl glycosides

Aquatic invertebrates:

EC50 (48 h) > 100 mg/l, *Daphnia magna* (Directive 92/69/EEC, C.2, static)

Nominal concentration.

LC50 (10 d) 3318 mg a.i./kg dw, *Corophium volutator* (, static)

Nominal concentration.

Information on: D-Glucopyranose, oligomeric, C10-16-alkyl glycosides

Aquatic plants:

EC50 (72 h) 12,5 mg/L a.i. (growth rate), *Desmodesmus subspicatus* (OECD Guideline 201, static)

Nominal concentration.

EC10 (72 h) 4,15 mg a.i./L (growth rate), *Pseudokirchneriella subcapitata* (OECD Guideline 201, static)

Nominal concentration.

Information on: D-Glucopyranose, oligomers, decyl octyl glycosides

Aquatic plants:

EC50 (72 h) 27,2 mg/L a.i. (growth rate), *Desmodesmus subspicatus* (OECD Guideline 201, static)

Nominal concentration.

Information on: D-Glucopyranose, oligomeric, C10-16-alkyl glycosides

Chronic toxicity to fish:

No observed effect concentration (28 d) 1,8 mg/l a.i., *Brachydanio rerio* (OECD Guideline 204, Flow through.)

Nominal concentration.

Information on: D-Glucopyranose, oligomers, decyl octyl glycosides

Chronic toxicity to fish:

No observed effect concentration (28 d) 1,8 mg/l a.i., *Brachydanio rerio* (OECD Guideline 204, Flow through.)

Nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)

Date / Revised: 06.10.2023

Version: 1.0

Product: **Plantatex® LLE Plus**

(ID no. 30843601/SDS_GEN_00/EN)

Date of print 08.02.2024

Information on: D-Glucopyranose, oligomeric, C10-16-alkyl glycosides

Chronic toxicity to aquatic invertebrates:

EC10 (21 d) 1,76 mg/l, Daphnia magna (OECD Guideline 202, part 2, semistatic)

Nominal concentration.

Information on: D-Glucopyranose, oligomers, decyl octyl glycosides

Chronic toxicity to aquatic invertebrates:

No observed effect concentration (21 d) 1,76 mg/l, Daphnia magna (OECD Guideline 202, part 2, semistatic)

Nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Analogous: Assessment derived from products with similar chemical character.

Assessment of terrestrial toxicity:

No data available concerning terrestrial toxicity.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):

Readily biodegradable (according to OECD criteria).

Information on: D-Glucopyranose, oligomeric, C10-16-alkyl glycosides

Assessment biodegradation and elimination (H₂O):

Readily biodegradable (according to OECD criteria).

Information on: D-Glucopyranose, oligomers, decyl octyl glycosides

Assessment biodegradation and elimination (H₂O):

Readily biodegradable (according to OECD criteria).

Information on: Citric acid

Assessment biodegradation and elimination (H₂O):

Readily biodegradable (according to OECD criteria). Easily eliminated from water.

Bioaccumulative potential

Assessment bioaccumulation potential:

Accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments:

Volatility: The substance will not evaporate into the atmosphere from the water surface.

Adsorption in soil: Adsorption to solid soil phase is possible.

Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative).

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Add. remarks environm. fate & pathway:

Treatment in biological waste water treatment plants has to be performed according to local and administrative regulations.

Other ecotoxicological advice:

The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components.

Due to the pH-value of the product, neutralization is generally required before discharging sewage into treatment plants.

13. Disposal Considerations

Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:

Uncontaminated packaging can be re-used.

Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport

ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel

Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information**Safety, health and environmental regulations/legislation specific for the substance or mixture**

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. This includes the mentioned and recommended usage. Any other intended applications should be discussed with the manufacturer. In particular this concerns the application for products that are the object of special standards and regulations.

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Eye Dam./Irrit.	Serious eye damage/eye irritation
Aquatic Acute	Hazardous to the aquatic environment - acute
Skin Corr./Irrit.	Skin corrosion/irritation
Acute Tox.	Acute toxicity
STOT SE	Specific target organ toxicity — single exposure
H318	Causes serious eye damage.
H315	Causes skin irritation.
H401	Toxic to aquatic life.
H402	Harmful to aquatic life.
H319	Causes serious eye irritation.
H303	May be harmful if swallowed.
H335	May cause respiratory irritation.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.